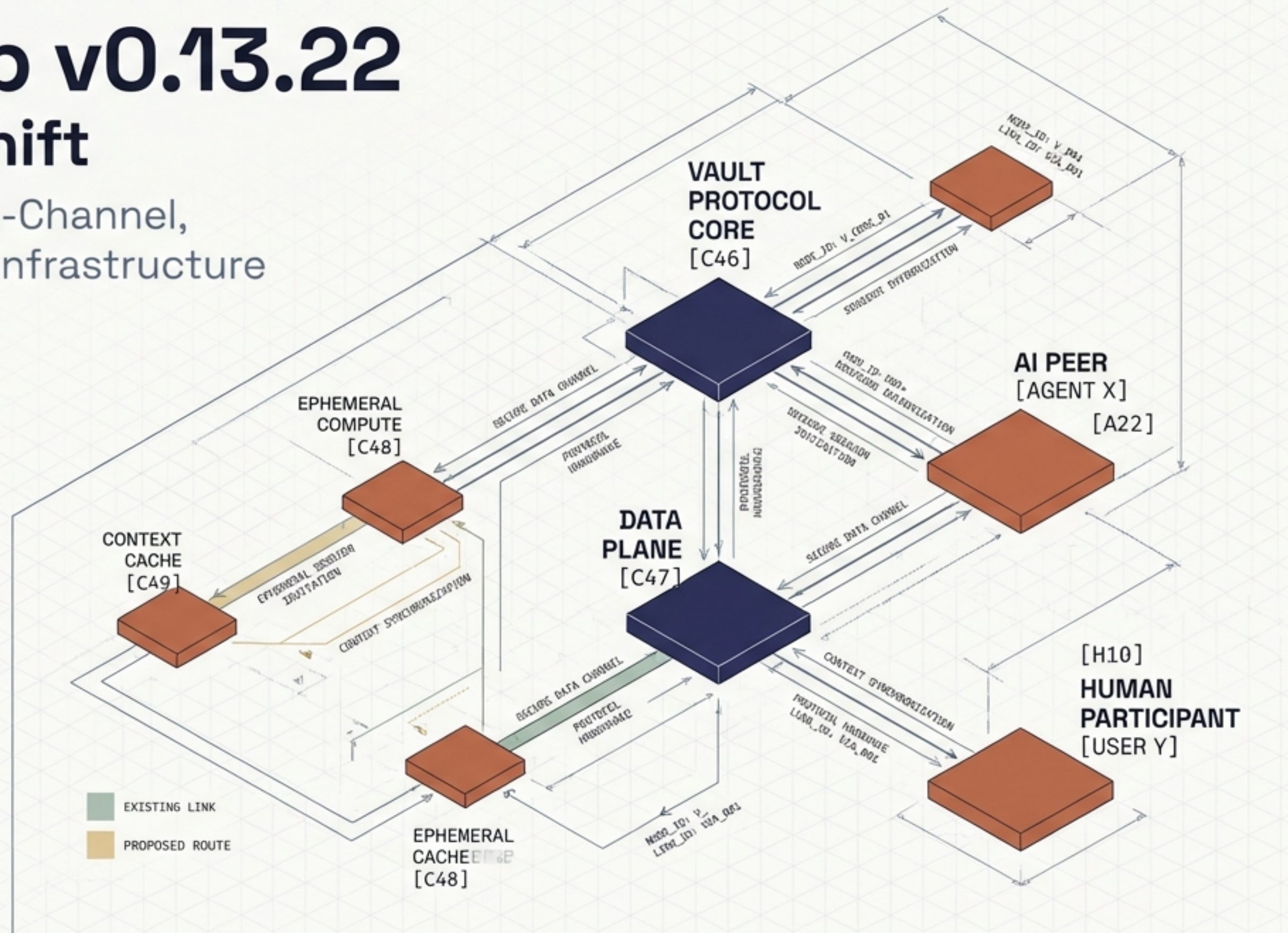


# System Map v0.13.22

## The Paradigm Shift

Vault-as-Communication-Channel,  
AI Peers, and Ephemeral Infrastructure



### Metadata Console

DATE: 11 March 2026

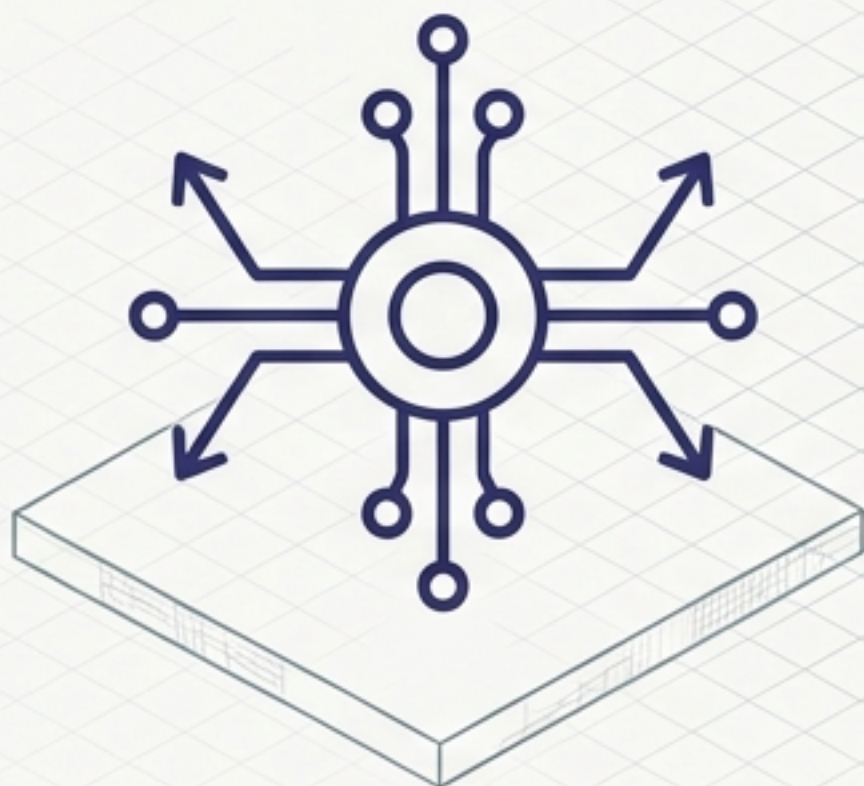
ROLE: Cartographer

TEAM: Explorer

STATUS: Round-trip proven

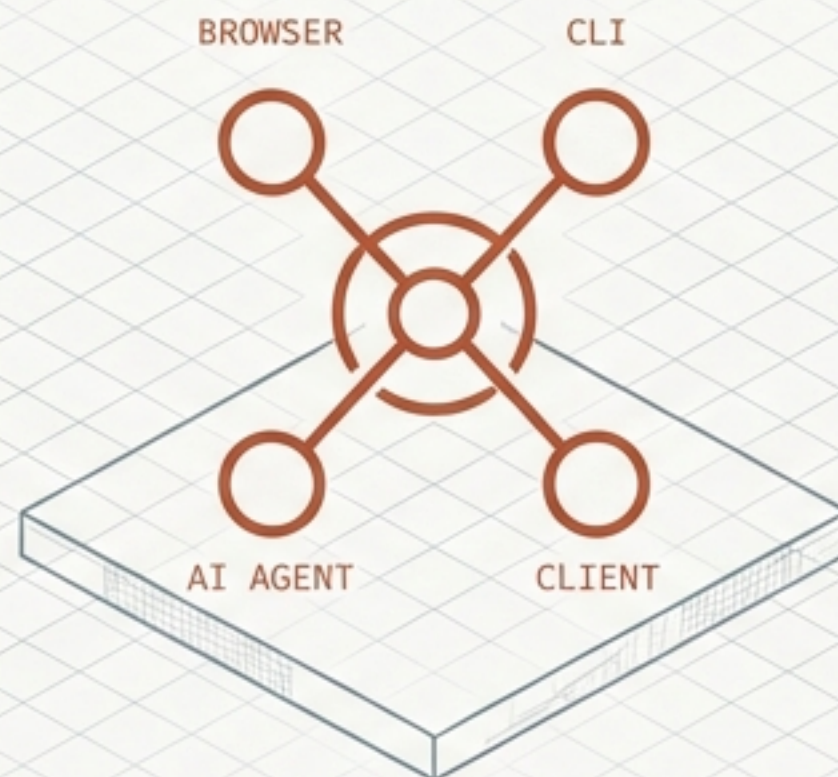


# Three architectural shifts redefine the system topology



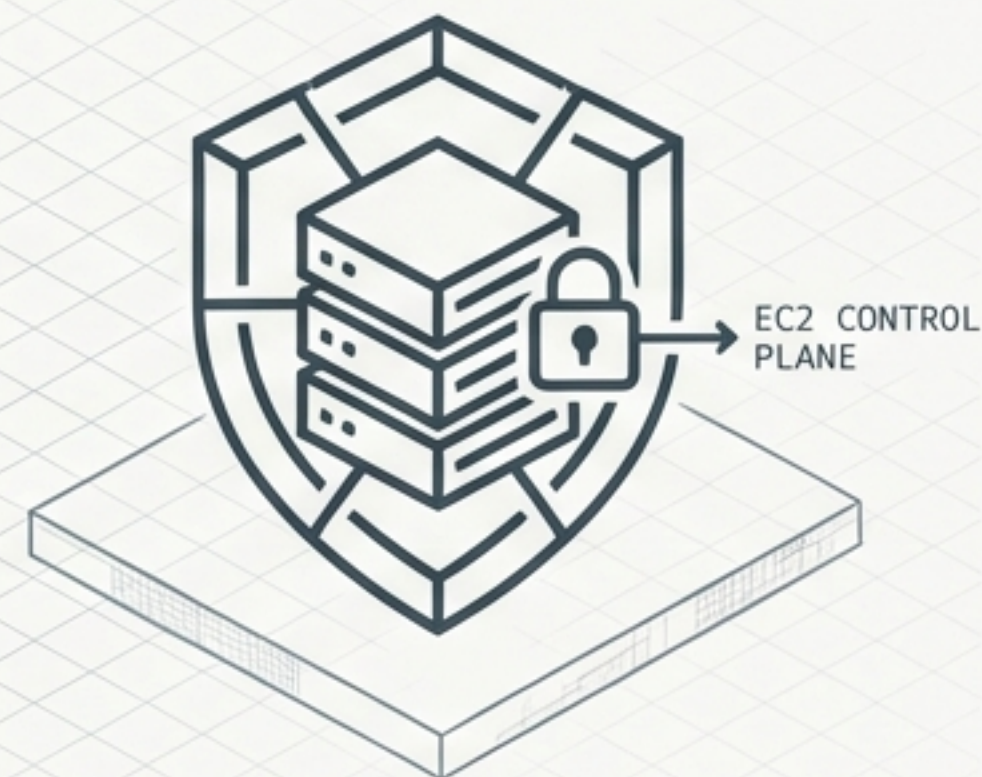
## Storage to Protocol

The vault has evolved beyond encrypted storage. It is now a bidirectional, asynchronous, and persistent persistent communication medium connecting humans and AI.



## Participant Expansion

The ecosystem has outgrown browser-only access. **The model now formally supports four participant types: Browsers, CLI, AI Agents, and Programmatic Clients.**



## Infrastructure as State

The proposed Deploy Service leverages the vault's zero-knowledge architecture to securely store state and SSH keys for ephemeral EC2 control planes.



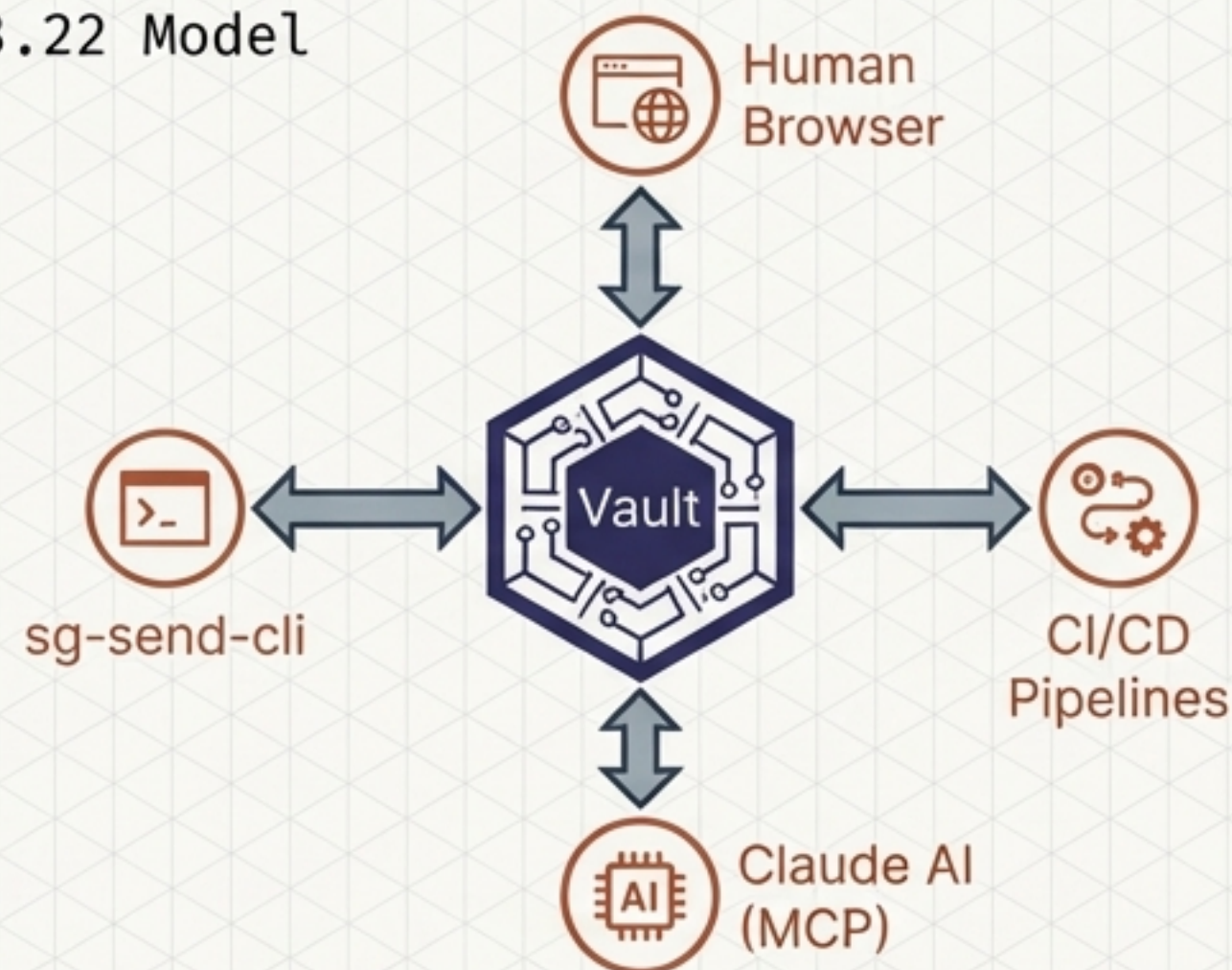
# The vault is no longer just storage—it is a communication protocol

v0.1.2 Model



Point-to-Point Delivery

v0.13.22 Model



Persistent Multi-Party Channel

File transfers are point-to-point delivery. Vaults are persistent, versioned, multi-party communication channels.

[C46: Vault-as-Communication-Channel]

[Status: EXISTS]

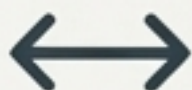
Full round-trip proven: Human → vault → Claude → vault → Human.



# Eight pillars of the new vault protocol

[EXISTS]

## Bidirectional



Both parties can actively read and write.

[EXISTS]

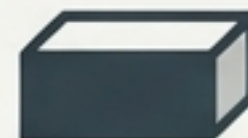
## Asynchronous



Parties do not need to be online simultaneously.

[EXISTS]

## Persistent



Vault context survives individual session boundaries.

[EXISTS]

## Versioned



Every push creates a content-addressed commit.

[EXISTS]

## Encrypted



Client-side AES-256-GCM ensuring zero-knowledge.

[EXISTS]

## Identity-Agnostic



Server infrastructure remains blind to reader/writer identities.

[EXISTS]

## Self-Describing



Vaults can contain their own execution instructions (SKILL.md).

[EXISTS]

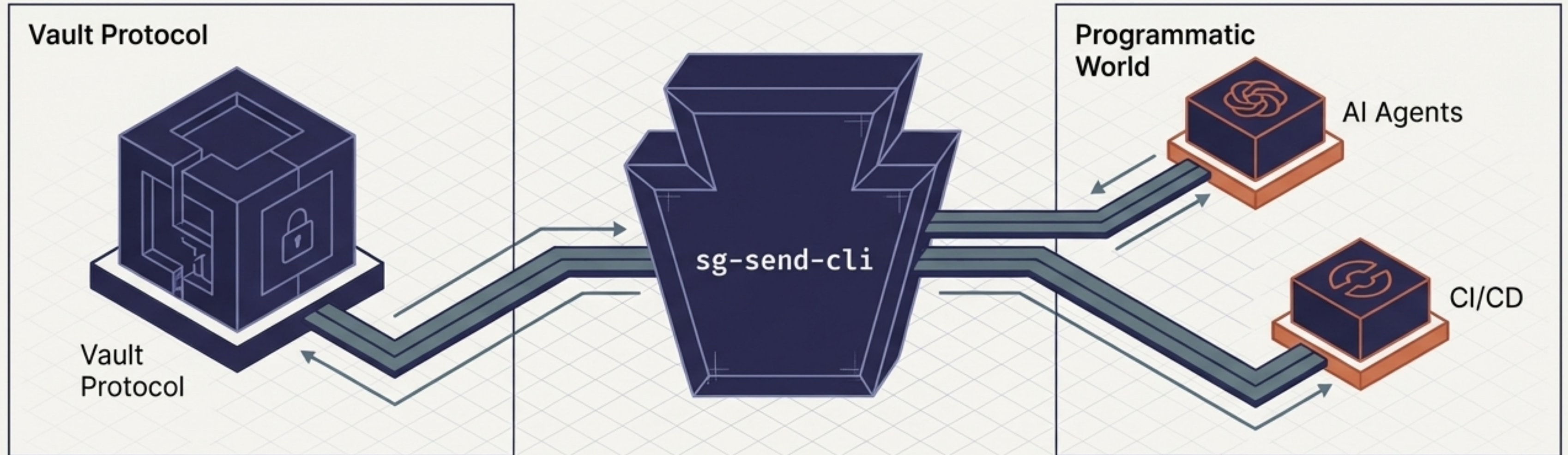
## Access-Controlled



Read requires vault key;  
Write requires vault key + token.



# sg-send-cli bridges the gap to programmatic peers



## Context:

Without the CLI, Claude sessions are limited to basic MCP tool calls. With it, they upgrade to full vault peers capable of complex operations.

[C47] Custom-Built, early. Currently available as v0.5.0 on PyPI.

## Working (Proven)

Vault Init, Clone, Push, Pull.

## Proposed Expansions

PKI operations, Secrets management (bypassing OS keychains), Upload/Download endpoints.



# Expanded capabilities reveal AI as the ultimate system peer

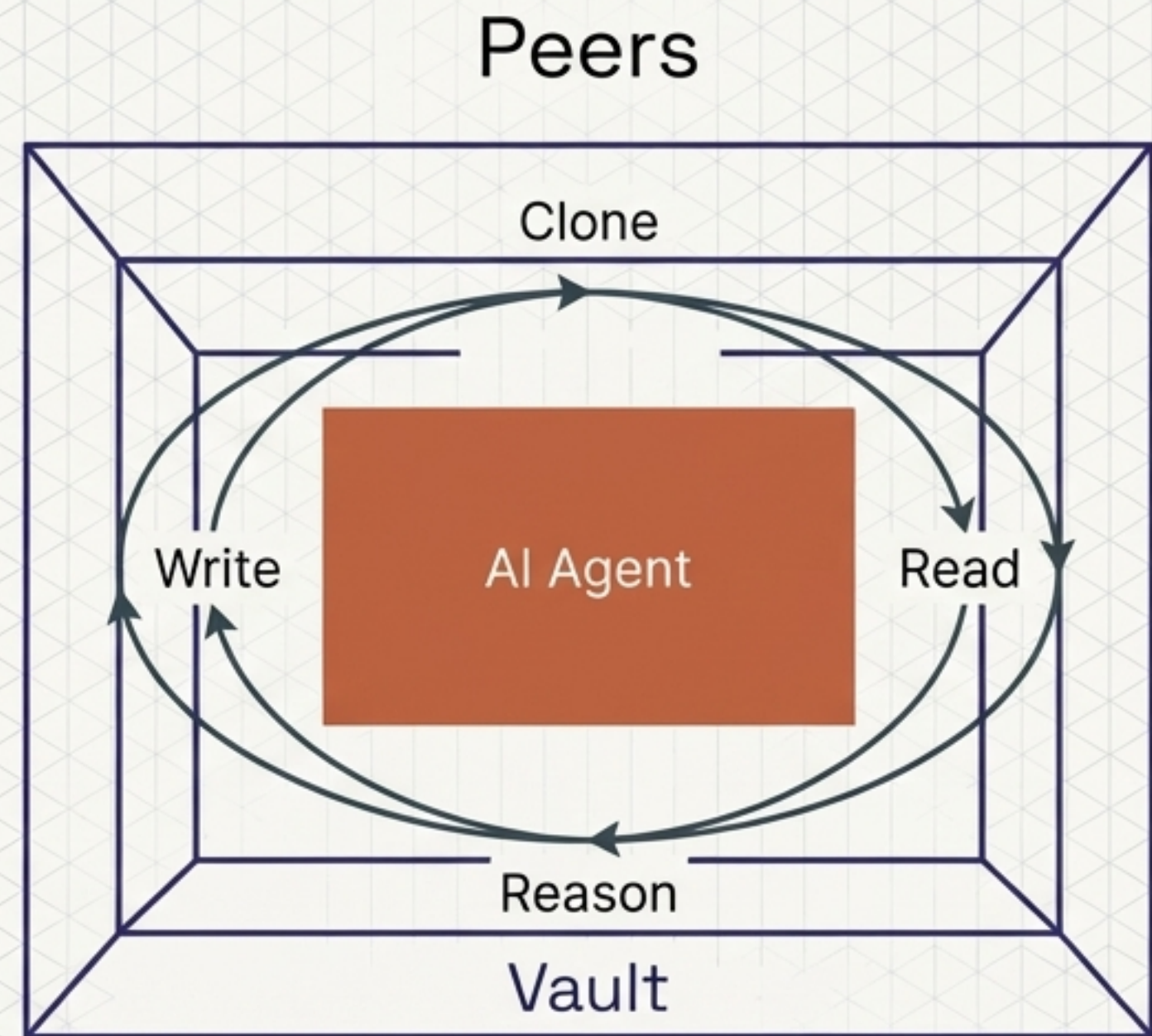
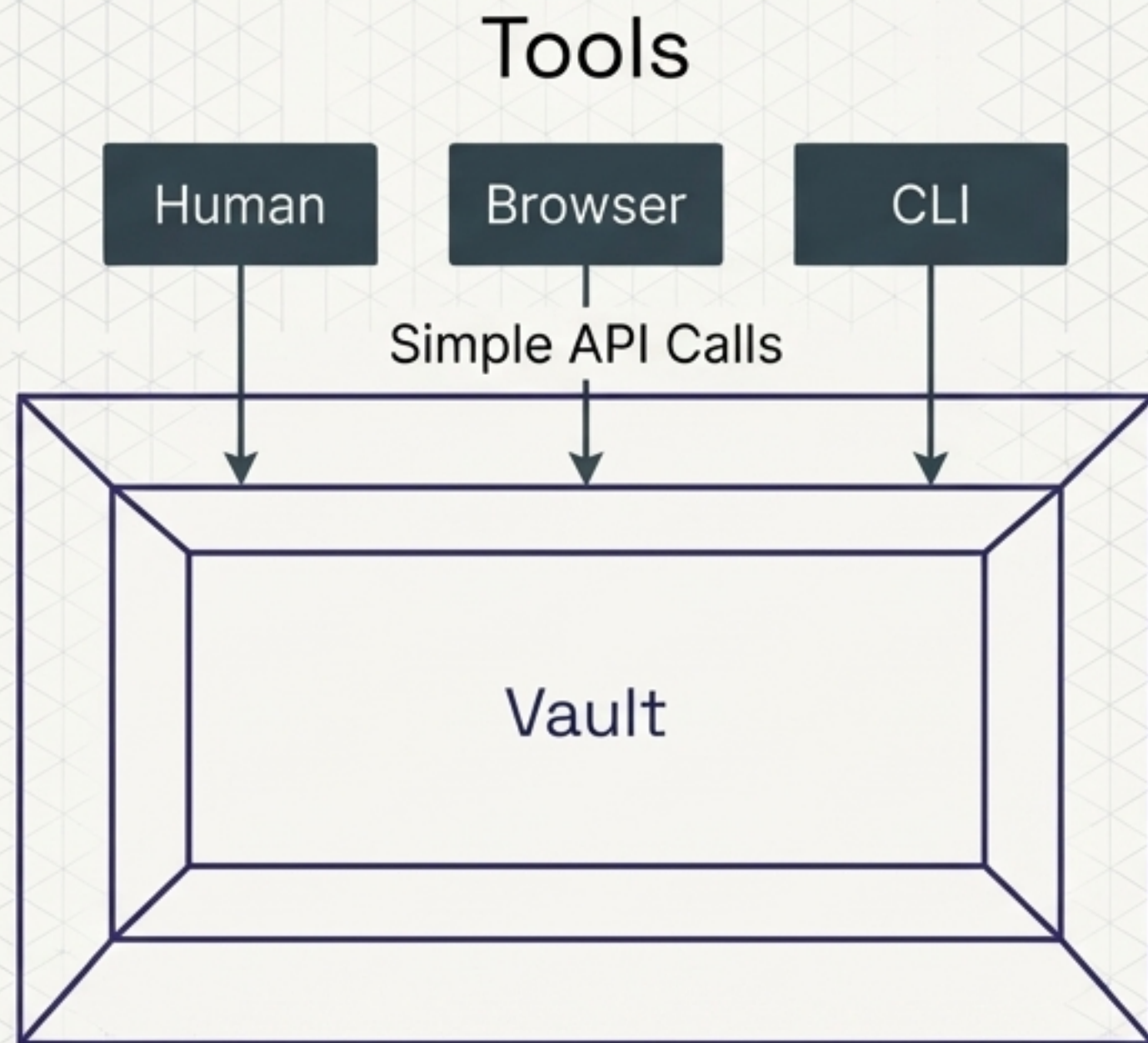
|   |                               | Browser | sg-send-cli | AI Agent | Programmatic       |
|---|-------------------------------|---------|-------------|----------|--------------------|
| 1 | Upload/Download               | ●       | ●           | ●        | ●                  |
| 2 | Init/Clone/Push/Pull          |         | ●           | ●        | ●                  |
| 3 | MCP Tools                     |         |             | ●        |                    |
| 4 | PKI & Secrets (PROPOSED)      |         | ○           | ○        | ○                  |
| 5 | Self-bootstrapping (SKILL.md) |         |             | ●        | Superpower<br>Zone |
| 6 | Reason about content          |         |             | ●        |                    |

Solid circle = EXISTS. Hollow circle = PROPOSED.

Browsers and CLIs are tools operated by external actors. AI agents are **the only participants that can natively read, parse, and reason about vault state.**



# Operating within the vault context, not just on it

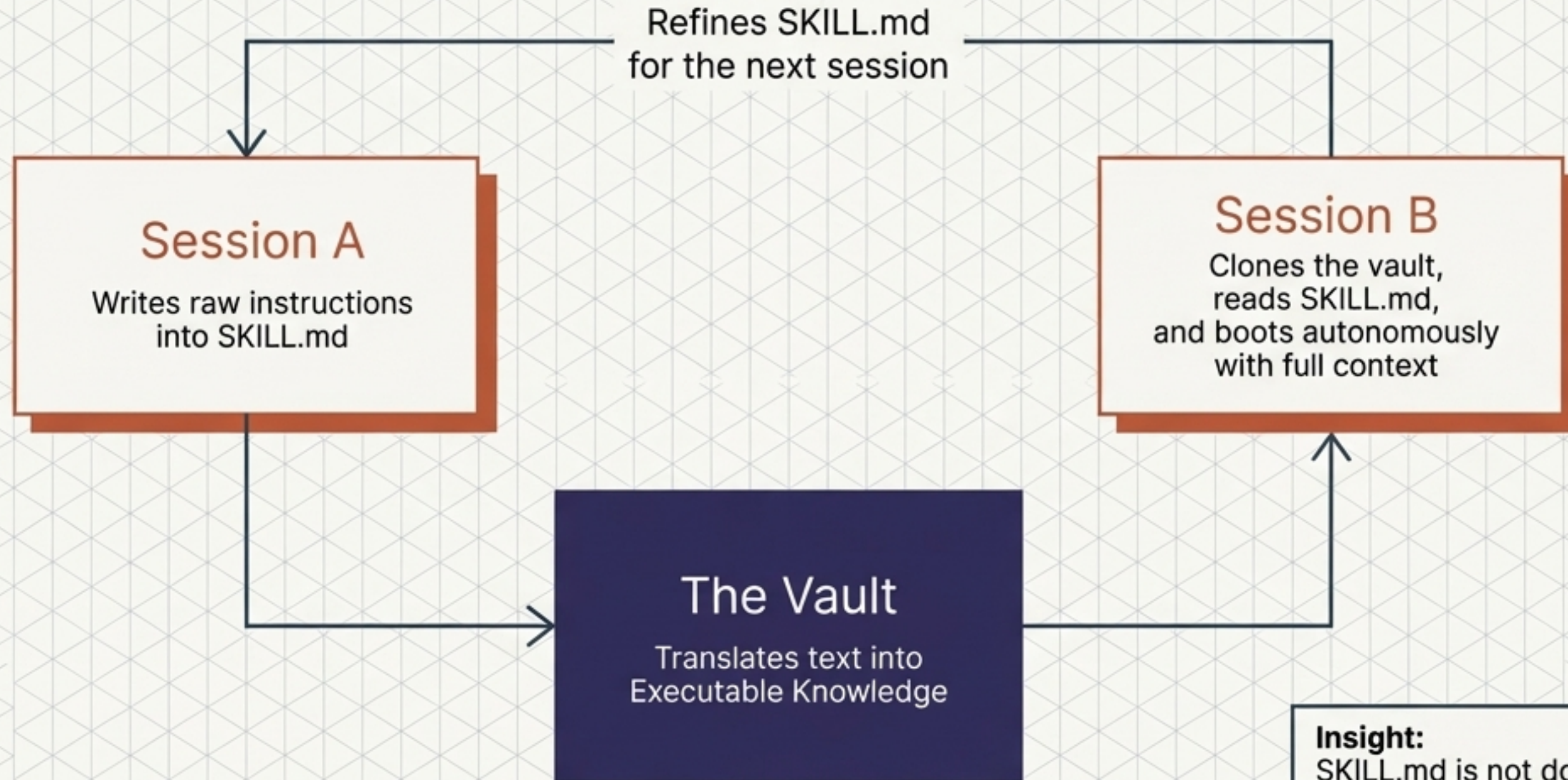


**Evolution Update:** Component [C48: AI Agent as Vault Peer] officially supersedes the legacy [C24: AI-as-sender via MCP].

Proven across two independent Claude sessions. AI agents do not just move files—they ingest the vault's history, understand its intent, and synthesize entirely new commits.



# Self-bootstrapping turns knowledge into executable state



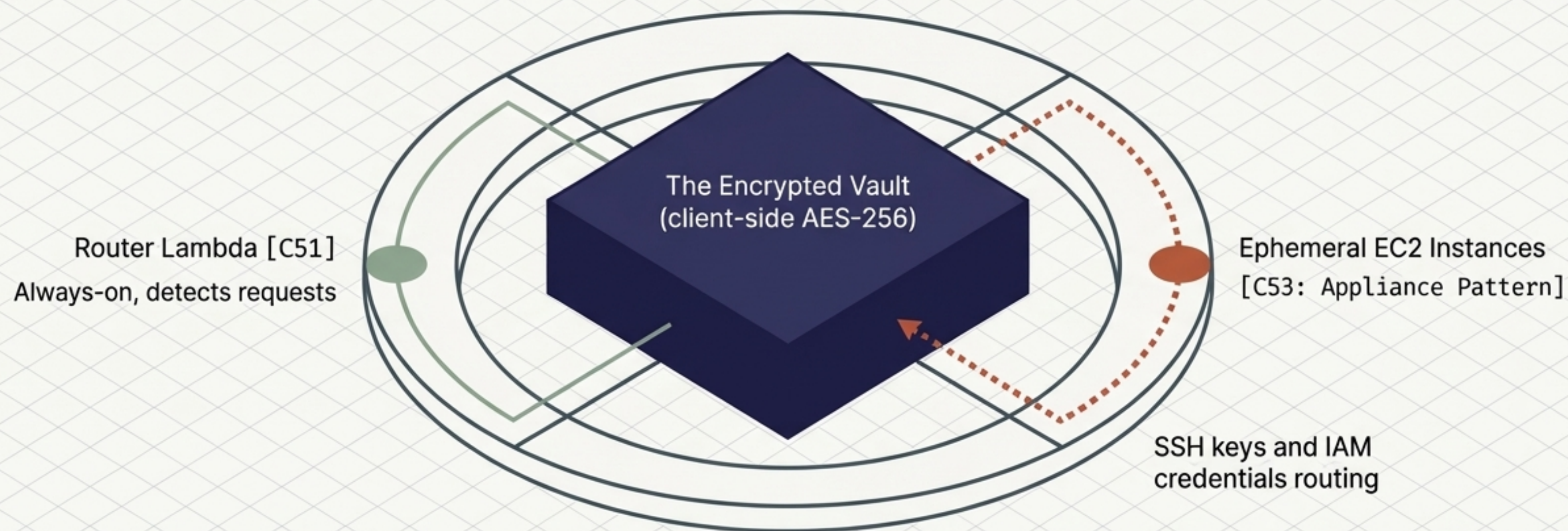
Definition:  
**Self-Bootstrapping**  
[Component C49]

**Insight:**  
SKILL.md is not documentation. It is executable state. It allows one ephemeral AI session to program the operating parameters of a future, independent AI session solely through the vault protocol.



# Applying zero-knowledge state to ephemeral infrastructure

## [Deploy Service: PROPOSED]



### The Concept

Position SG/Send's vault as the state store for ephemeral cloud infrastructure.

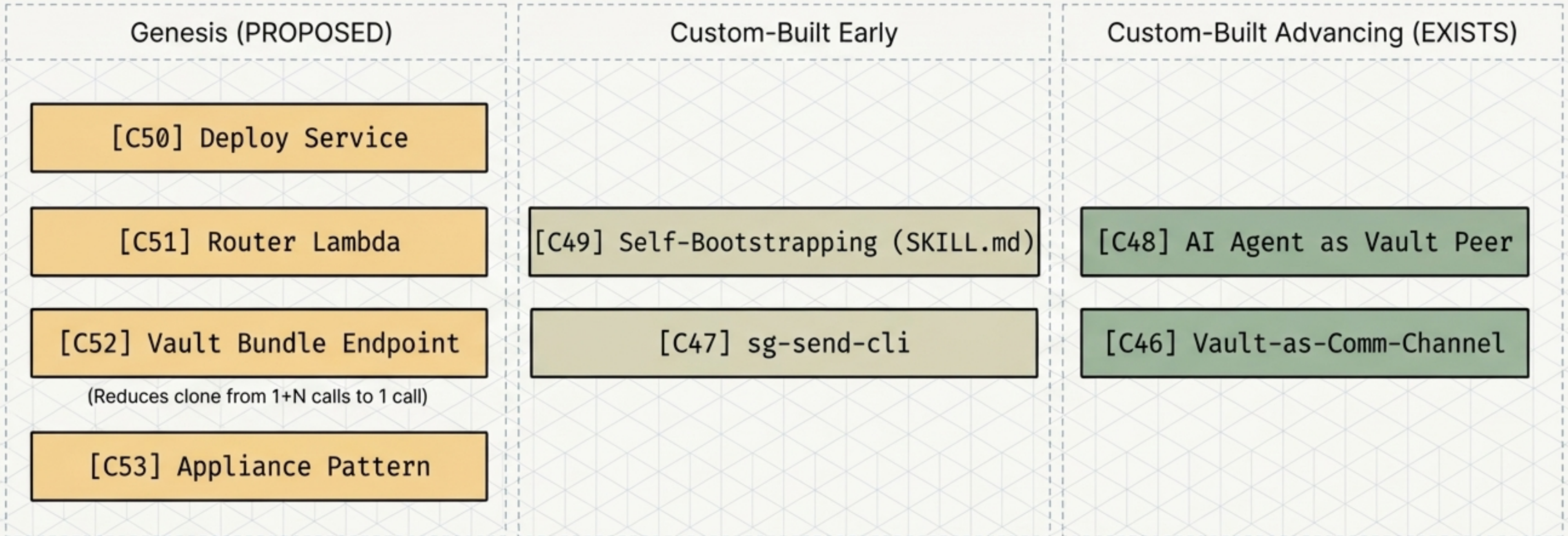
### The Security Guarantee

Because the vault enforces zero-knowledge, the central SG/Send server never sees the infrastructure SSH keys or IAM credentials. State remains completely dark to the intermediary.



# Tracking component momentum across the March 10 sprint

## Evolution Stage



The overall Vault System has shifted rapidly rightward, backed by 71+ tests and proven round-trip execution.



# Architectural axioms for the new communication paradigm



## Infrastructure, Not A Feature

The vault is no longer a utility within the system; it is the foundational communication primitive connecting all human, AI, and hardware actors.



## Participants > Features

System value is now determined by who can connect and how they communicate, driving the four-participant model as the primary axis of expansion.



## The CLI is the Gatekeeper

`sg-send-cli` dictates the capability surface for all non-browser participants. It is the mandatory bridge for AI and CI/CD integration.



## Knowledge is Executable

Through self-bootstrapping (`SKILL.md`), AI sessions can iteratively improve the vault's instructions, creating autonomous, cross-session feedback loops.